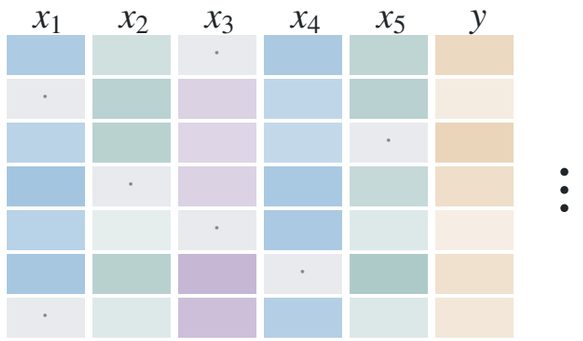
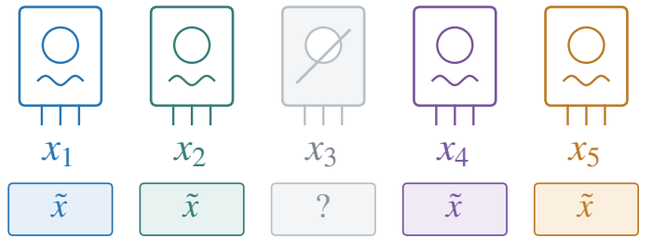
(a) Deployment calibration

Support $\mathcal{D} = \{(x_i, m_i, y_i)\}_{i=1}^{48}$



Illustrative cells, 48 rows cropped

(b) Query sensor subset

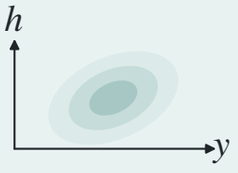


$S = \{1, 2, 4, 5\}$; sensor 3 is absent

Lift the latent state

$z = (y, h) \in \mathbb{R}^{1+K}$

Target and shared nuisance



(c) Fit the Gaussian anchor

Incomplete support only

Gaussian EM, then factor analysis

$A,\ D,\ \psi,\ \mu_x,\ \mu_y,\ v_y$

(d) Choose a learned site route

Lifted cavity network

Shared f_θ; $R = 3$ cavity updates

or

Lifted cavity transformer

Query cell head; one site update

(e) One site per observed sensor

$t_j(z) \longleftrightarrow (\Lambda_j, \eta_j)$

1 $w_j,\ o_j,\ \psi_j$ Λ_j η_j

2 $w_j,\ o_j,\ \psi_j$ Λ_j η_j

3 Absent sensor: no factor

4 $w_j,\ o_j,\ \psi_j$ Λ_j η_j

5 $w_j,\ o_j,\ \psi_j$ Λ_j η_j

$$\Lambda_j = \frac{w_j w_j^\top}{\psi_j}, \qquad \eta_j = \frac{w_j o_j}{\psi_j}$$

Rank one sites over the same $z = (y, h)$

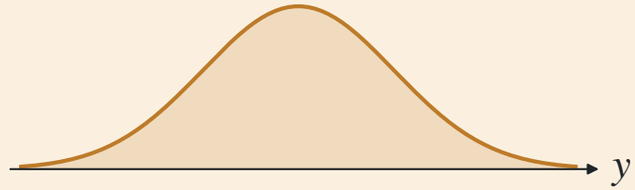
(f) Add natural parameters and marginalize

$$\Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j$$

$$\eta = \eta_0 + \sum_{j \in S} \eta_j$$

$$m = \Lambda^{-1}\eta, \qquad V = \Lambda^{-1}$$

$$q(y) = \mathcal{N}(m_y, V_{yy})$$



Predictive target distribution